Bernardo Mondragón Brozon

Lead Full-Stack Engineer | Data & Cloud Architect | AI/ML Practitioner 🌐 bernardomondragon.net | GitHub.com/BernardoMB 📧 bmondragonbrozon@gmail.com | **Bachelor's in Actuarial Science**

PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with **14+ years** building large-scale, enterprise, data-driven and cloud-native systems. Expert in **.NET, Node.js, Angular, React, Azure, SQL, Databricks, Spark**, and modern architectural patterns (DDD, event-driven, microservices, serverless). Strong background in **AI/ML, risk modeling, insurance systems, and high-impact data platforms**.

Proven track record delivering systems used by **100k+ users**, processing **millions of records**, and supporting mission-critical operations in **government, insurance, logistics, and enterprise environments**. Experienced in **architecture, leadership, mentoring, and end-to-end delivery**.

CORE SKILLS

Backend: .NET (C#), Node.js, Python, Microservices, REST, SOAP, XML, TCP/IP
Frontend: Angular, React, TypeScript, Blazor, ASP.NET, Vue
Cloud: Azure, AWS, GCP, Vercel, Containerization & Orchestration, Serverless Design, Event Driven Architectures, Power BI, Docker, CI/CD , Terraform, Observability & Reliability CloudWatch, Prometheus/Grafana, distributed tracing.
Data: SQL Server, PostgreSQL, MongoDB, Databricks, Spark, Delta Lake, Power BI, Cosmos DB.
AI/ML: Python, Machine Learning, scikit-learn, Intelligent Document Processing, Deep Learning, LLMs, forecasting models, Agentic Development, Big data, EDA, Machine learning, Azure ML, RAG.
Other: DDD, distributed systems, event-driven architecture, Serverless Design, DevOps, Agile/Scrum, High Volume & High Throughput, Embeddings, Cursor, Claude Code

PROFESSIONAL EXPERIENCE

Senior Developer — Toyota Material Handling (TMHNA)

Columbus, IN | Mar 2022 – Present

> **Tech Stack:** ASP.NET, .NET, Razor Pages, Blazor, Telerik, Kendo UI, Node.js, Angular, SQL Server, Azure (App Service, Key Vault, Azure SQL, App Configuration Service, Functions, Blob, Active Directory, OAuth2, Delta Lake, AI Search, Redis, APIM, ADF, Azure Events), Power BI, Azure DevOps, Azure DevOps Pipelines, Azure Insights, Cosmos DB, Serverless, Intelligent Document Processing, Agentic Development, Agentic PDLC, AI Orchestration, C#, TypeScript, JavaScript, HTML, CSS, SCSS, SQL, T-SQL, KQL, DAX, Power Query M, Python, PowerShell, Bash, YAML, JSON, VB.NET, F#

Key Achievements

- Standardized financial processes across IT.
- Reduced manual workload through automation.
- Improved decision-making with real-time analytics and dashboards.

Initiatives

Go-To-Market Tools

- Internal enterprise platform used by **100+ users** for pricing, cost management, forecasting, reporting, and analytics.
- Generated **thousands of quotes** and automated discount approvals.
- Integrated **Azure AI Search**, Redis caching, and distributed processing for high performance.

Houkoku Renraku Soudan (HRS)

- Full IT Asset Management system aligned with **auditor and governance controls**.
- Automated **Asset Expenditure Requests, SOWs, and invoicing** using AI-based document processing.
- Improved financial transparency and stakeholder trust.

Mercury

- Pricing-approval and analytics tool for analysts and executives.
- Added dashboards, trend analysis, embedded Power BI, and approval workflows.

Senior Full-Stack Developer — OSSE (Government of Washington DC)

Washington, DC | Mar 2022 – Mar 2023

> **Tech Stack:** Angular, .NET, SQL Server, Azure Functions, Blob Storage, Redis, AAD, B2C, GitHub Actions, Pipelines, Excel Integrations, PDF Generation, Entity Framework, Python, SharePoint, TypeScript, JavaScript, HTML, CSS, SCSS, C#, SQL, T-SQL, PowerShell, Bash, YAML, JSON

Key Achievements

- Delivered mission-critical systems for public safety, health, and education.
- Enabled real-time reporting for the Mayor, DC Council, and the Healthy Youth & Schools Commission.
- Improved the lives of 10000+ children helping their school adopt better health standards
- Helped District's schools to achieve the goals of the Healthy Schools Act of 2010
  Made 251 schools healthier and made a positive impact on 93,843+ students
- Promptly and properly notified and informed schools, parents, and emergency services about incidents, accidents, and complaints

Initiatives

Division of Transportation Case Management System

- Managed and maintained **200,000+ cases** for student transportation.
- Improved transportation services for **93,843+ students** across 251 schools.
- Built dynamic forms, notifications, PDF generation, and integrations with multiple external systems.

School Health Profiles Portal

- Developed a dynamic, metadata-driven form engine supporting 400+ questions to assess school health compliance.
- Enabled compliance reporting for the **Healthy Schools Act**, used by **1,000+ users**.
- Built dashboards, reports, and data-access layers for LEAs and government officials.

Data Scientist — One Auction View (OAV)

Redlands, CA | Aug 2018 – Aug 2019

> **Tech Stack:** React, Node.js, NestJS, Cosmos DB, Databricks, Python, Spark, Power BI, Azure Functions, PostgreSQL, Docker, Terraform, AI, Cypress, Playwright, AI Vibe Coding, TypeScript, JavaScript, HTML, CSS, SQL, C#, YAML, JSON, DAX, Power Query M, Bash, PowerShell

Initiatives

Retail Vision by One Auction View (Data Platform)

- Processed and analyzed **multi-million-record datasets** from the automotive industry.
- Built dashboards and forecasting models used by **20+ dealerships** and financial institutions.
- Designed and deployed a nationwide recommendations engine for U.S. vehicle dealerships, enabling data-driven inventory acquisition to maximize profitability, fully integrated with One Auction View to streamline the end-to-end recommendation-to-purchase workflow.
- Designed data pipelines using **Databricks, Delta Lake, and distributed computing**.
- Improved inventory decisions and profitability through segmentation and predictive analytics.

OAV Portal (Head of QA)

- Built a unified auto auction platform for dealerships.
- Developed a multi-auction vehicle sourcing platform.
- Created a centralized vehicle acquisition portal.
- Built a dealer inventory acquisition platform.
- Developed a vehicle sourcing and auction management portal.

Full-Stack Developer — Zurich Insurance Company

Mexico City | Feb 2020 – Mar 2022

> **Tech Stack:** .NET, Blazor Server, SQL Server, Azure, Redis, Jenkins, Jira, Cosmos DB, Docker, Blazor Web Assembly, Microsoft Entity Framework, Dynatrace, Azure Pipelines, Azure DevOps, Third Party Integrations, SharePoint, Office SDKs, R, Python, BigQuery, BigQuery ML, Vertex AI, Snowflake, Snowpark, statsmodels, scikit-learn, PySpark, XGBoost, LightGBM, Docker, Google Cloud Platform, Cloud Storage, Cloud IAM, Cloud DLP, Cloud Scheduler, Vertex AI Pipelines, Kubeflow, Vertex AI Vizier, Vertex AI Model Registry, Looker Studio, BigQuery BI Engine, REST APIs, C#, JavaScript, TypeScript, HTML, CSS, SQL, T-SQL, PowerShell, Bash, YAML, JSON

Key Achievements

- Enabled financial Analysts, Underwriters, and Account Managers to automate their workflows.
- Successfully developed and deployed sureties administration system for Zurich North America.
- Successfully developed and deployed underwriting systems for exporter's property and casualty in North America.

Initiatives

Sureties ERP (North America)

- Built underwriting automation, financial analysis tools, and policy management workflows.
- Migrated legacy systems while maintaining production stability.
- Implemented risk assessment, pricing, financial statements parsing, and compliance modules.

Exporter Solutions

- Customer-facing underwriting and policy-issuance platform used by **10,000+ brokers**.
- Delivered full workflow: data entry → pricing → underwriting → policy issuance.
- Data engineering: Built an end-to-end pricing pipeline on the CAS insurance dataset. R handles ingestion into partitioned BigQuery tables, SQL and BigQuery ML handle cleaning, standardization and feature engineering, and the features are kept in reusable feature tables.
- Actuarial modeling at scale: Trained Poisson/Negative Binomial frequency and Gamma/Lognormal severity GLMs in Python (statsmodels / scikit-learn), and benchmarked them against XGBoost/LightGBM using SHAP for interpretability. Training ran as containerized Vertex AI custom jobs, orchestrated with Vertex AI Pipelines and tuned with Vizier.
- Production MLOps and governance: Served premium predictions through a REST endpoint, with drift and fairness monitoring and automated retraining via Cloud Scheduler. Versioned models in the Model Registry and secured data with IAM and Cloud DLP to meet regulatory requirements. Costs stayed under $35 through right-sized, one-off compute.

Co-Founder — Kalyptio

Mexico City | Dec 2017 – Jan 2021

> **Tech Stack:** Amazon RDS for SQL Server, AWS Elastic Beanstalk, AWS Secrets Manager, AWS KMS, AWS AppConfig, AWS Lambda, Amazon S3, AWS IAM Identity Center, AWS Directory Service, Amazon Cognito, AWS Lake Formation, Amazon API Gateway, AWS Glue, AWS Step Functions, Amazon EventBridge, Amazon CloudWatch, AWS X-Ray, Amazon DynamoDB, Amazon CloudWatch Logs Insights, Amazon EMR, Amazon RDS for PostgreSQL, Amazon Aurora, Amazon ECS, Amazon ECR,

> AWS Fargate, AWS CloudFormation, AWS CDK, Amazon SageMaker AI, Amazon RDS for MySQL, Amazon EC2, AWS Outposts, Amazon Redshift, AWS IAM, Terraform

Led strategy, architecture, UX, development, DevOps, and delivery for **30+ clients**.

Initiatives

- **Aurora Admin Panel:** CMS for courses, events, news, and users (Angular + NestJS + MongoDB).
- **Staff Plan:** Staff management system (AWS + Lambda + Node.js).
- **Eva:** Marketing Funnel (AWS + Lambda + Python).
- **Medkush:** E-commerce platform (Angular + Serverless + MongoDB).
- **Newt Payments:** Payments Platform (Angular + Nest + MongoDB + SQL + AWS).
- **Finero:** Payments and remittances platform (Angular + Node.js + AWS).

Lead Developer — Neolinx

Mexico City | Jan 2012 – Nov 2017

> **Tech Stack:** ASP.NET, .NET, Ext.NET, Angular, Node.js, SQL Server, Oracle, SignalR, Firebase, Azure, MySQL, Java, FTP, Oracle DB, PL/SQL, Linux Servers, On-prem Deployments, D3.js, Angular Material, Postman, SOAP, TCP/IP, C#, TypeScript, JavaScript, HTML, CSS, SCSS, SQL, T-SQL, PowerShell, Bash, YAML, JSON

Key Achievements

- Processed **700M+ telecom records (CDRs)** for national security agencies (PGR, CISEN).
- Built real-time analysis tools used in tactical operations, helping prevent crimes and save lives.
- Delivered full-stack systems for link analysis, geolocation, and intelligence workflows.

Initiatives

Matrix Catcher

- Locate mobile phone subscribers covertly anywhere in the world using telecom infrastructure rather than fake cell towers.
- Determine status, location, and movement of targets across a city, region, or entire country with high precision.
- Perform massive numbers of searches — e.g., 135,000 real-time location lookups under a single contract.

Trinity

- Full call-content interception: RCS allowed authorities to listen to live phone calls, record them, and capture VoIP calls (WhatsApp, Skype, Viber) directly from infected devices.
- Device-level infection for total access: The system infiltrated phones through remote exploits or physical access, turning the device into a surveillance sensor capable of capturing calls, ambient audio, messages, files, and more.
- Operating platform: Maintained the system for Mexican agencies, providing infrastructure and operational support, leveraging interception technology developed by Hacking Team (Italy).

Morpheus

- Central surveillance dashboard: Morpheus functioned as Neolinx's unified command-and-control interface, letting Mexican agencies manage targets, organize cases, and view results from multiple interception systems in one place.
- Data-fusion and orchestration layer: It aggregated outputs from foreign tools Neolinx sold — Hacking Team RCS, SS7/telecom lawful-interception modules, Rayzone geolocation and metadata platforms — without performing interception itself.
- Operational workflow system: Morpheus provided automation for surveillance tasks (tasking, evidence tagging, reporting, chain-of-custody), serving as the operational glue that connected and streamlined all other SIGINT tools Neolinx deployed

Zion

- Real-time operations console: Zion served as Neolinx's live monitoring interface, showing active infections, geolocation queries, and lawful-interception streams in one unified dashboard.
- Multi-system orchestration: It coordinated actions across all surveillance tools Neolinx deployed—triggering tasks, pulling data, and synchronizing results—without performing interception itself.
- Audit and compliance layer: Zion logged operator actions, maintained chain-of-custody records, and packaged evidence for legal or internal review, acting as the oversight framework for surveillance operations.

EDUCATION

Bachelor's in Actuarial Science — *ITAM (2013–2018)* Applied mathematics, statistics, risk modeling, insurance theory.

Insurance Diploma — *ITAM (2020–2021)* Insurance Theory in Depth for types of Insurance and how insurance companies operate.

Solvency II Diploma — *UNAM (2019)* Diploma on Solvency II regulation for Insurance companies in Mexico.

KEY ACHIEVEMENTS

- Built systems used by **100k+ users** across government and enterprise.
- Processed **700M+ records** for national security intelligence.
- Delivered platforms managing **200k+ cases** and improving services for **93k+ students**.
- Designed data pipelines processing **millions of automotive records**.
- Led architecture and delivery for **30+ commercial clients**.